

ENTHUSIAST ADVANCE COURSE

LEADER
PHASE - I (B)
TARGET : PRE-MEDICAL 2024

Test Type : **MINOR**

Test Pattern : **NEET (UG)**

TEST DATE : 04-03-2023

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	4	2	3	1	1	4	4	2	1	2	2	3	3	1	2	2	1	3	3	2	4	1	2	2	1	1	4	3	2	3
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	1	2	4	3	1	2	1	3	4	3	3	1	2	3	3	1	1	4	2	2	3	2	4	2	3	3	4	1	1	4
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	2	2	3	3	1	3	3	1	4	1	4	3	1	1	2	2	3	2	1	2	3	4	3	3	4	2	4	2	1	2
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	4	4	3	2	2	3	3	3	3	4	4	3	3	1	4	2	3	3	3	1	2	2	2	1	3	3	2	2	3	3
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	2	3	3	3	3	4	4	2	4	3	3	3	2	4	1	4	2	4	4	1	1	2	2	3	4	4	2	3	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	3	4	2	2	3	4	2	2	4	2	1	1	3	3	3	3	2	4	2	2	2	4	3	4	1	4	1	3	2	4
Q.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200										
A.	1	1	2	3	2	4	1	2	4	2	3	3	1	3	3	3	1	4	3	1										

HINT - SHEET

SUBJECT : PHYSICS

SECTION-A

1. **Ans (4)**

The time period of moon's revolution around the earth does not depend upon the mass of the moon.

2. **Ans (2)**

$$\frac{-GMm}{R} + \frac{1}{2}m\left(\frac{3}{4}V_e\right)^2 = -\frac{GMm}{R+h} + 0$$

$$\frac{-GM}{R} + \frac{1}{2} \times \frac{9}{16} \times \frac{2GM}{R} = -\frac{GM}{R+h}$$

$$\frac{-7GM}{16R} = -\frac{GM}{R+h}$$

$$7R + 7h = 16R$$

$$h = \frac{9R}{7}$$

3. **Ans (3)**

$$E = \frac{GMm}{2r}$$

$$L = mv_0r = m\sqrt{\frac{GM}{r}}r$$

$$L = \sqrt{\frac{GMm^2r^2}{r}}$$

$$L = \sqrt{\frac{2GMm}{2r}} \times mr^2$$

$$L = \sqrt{2Emr^2}$$

4. **Ans (1)**

$$a = \sqrt{\frac{R_1^2 + R_2^2}{m}} = \frac{R_3}{m}$$

5. **Ans (1)**

$$\text{Apparent weight} = m(g+a)$$

$$1.10 \times 10 = 1(10+a)$$

$$a = 1 \text{ m/s}^2$$

6. **Ans (4)**

$\Delta P = \text{area under the } F-t \text{ graph}$

$$\Delta P = \frac{1}{2} \times 8 \times 1 - 2 \times 0.5 = \frac{3\text{kg} - m}{s}$$

7. **Ans (4)**

$$T - 1 \times g = 1 \times 2a \dots\dots\dots(1)$$

$$4g - 2T = 4 \times a \dots\dots\dots(2)$$

$$8a = 2g$$

$$a = \frac{g}{4}$$

thus, the acceleration of the mass 1 kg is $2a$, so $\frac{g}{2}$ upwards

8. **Ans (2)**

$$\Delta U = U_f - U_i$$

$$= \left(\frac{-GMm}{4R} \right) - \left(\frac{-GMm}{R} \right)$$

$$= \frac{-GMm}{4R} + \frac{GMm}{R}$$

$$= \frac{3GMm}{4R}$$

$$\Delta U = \frac{3}{4} mgR \quad \left(g = \frac{GM}{R^2} \right)$$

9. **Ans (1)**

$$-\frac{3}{2} \frac{GMm}{R} + \frac{1}{2} mv^2 = -\frac{GMm}{R}$$

$$\Rightarrow v = \sqrt{\frac{GM}{R}}$$

10. **Ans (2)**

$$T^2 = \frac{4\pi^2 r^3}{GM}$$

$$\left(\frac{T_A}{T_B} \right)^2 = \left(\frac{r_1}{r_2} \right)^3$$

$$\left(\frac{T_A}{T_B} \right)^2 = \left(\frac{4}{1} \right)^3$$

$$\frac{T_A}{T_B} = 8$$

11. **Ans (2)**

Acceleration due to gravity at a height h from earth's surface

$$g_h = \frac{g}{\left(1 + \frac{h}{R} \right)^2}$$

$$\frac{g}{100} = \frac{g}{\left(1 + \frac{h}{R} \right)^2}$$

$$1 + \frac{h}{R} = 10$$

$$h = 9R$$

12. **Ans (3)**

$$T = \left(1 - \frac{x}{L} \right) F = \left(1 - \frac{1}{5} \right) = 5 = 4N$$

13. **Ans (3)**

$$T \cos 37^\circ = 20$$

$$T \left(\frac{4}{5} \right) = 20$$

$$T = 25 \text{ N}$$

14. **Ans (1)**

$$W_1 = m(g + a) = mg + ma$$

$$W_2 = m(g - a) = mg - ma$$

$$\text{So, } W_1 + W_2 = 2mg$$

$$\therefore W_1 + W_2 = 2W$$

$$\therefore W = \frac{W_1 + W_2}{2}$$

16. **Ans (2)**

$$a = \frac{(2+2)g - 2g}{2+2+2} = \frac{g}{3} = \frac{10}{3} \text{ m/s}^2$$

$$2 \times 10 - T = 2 \times \frac{10}{3}$$

$$\therefore T = 20 - \frac{20}{3} = \frac{40}{3} \text{ N}$$

$$T = 13.3 \text{ N}$$

18. **Ans (3)**

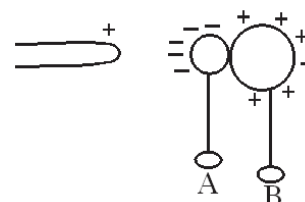
$$n = \frac{q}{e} = \frac{14.4 \times 10^{-19}}{1.6 \times 10^{-19}} = 9$$

Positive charge mean lost 9 electrons.

19. **Ans (3)**

When both spheres are in contact, charge

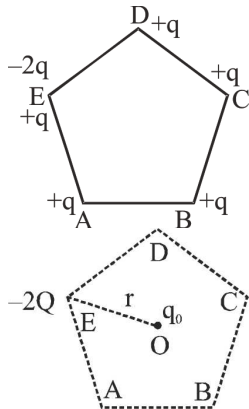
induced will be as below



Now, if sphere B is moved away, A has

negative & B has positive charge.

20. **Ans (2)**



Since, it is a regular polygon If all charges were positive net force would be zero. Now one positive is missing and one additional $-Q$ is placed at point E.

$$\therefore F_{\text{net}} = \frac{2kQq_0}{r^2}$$

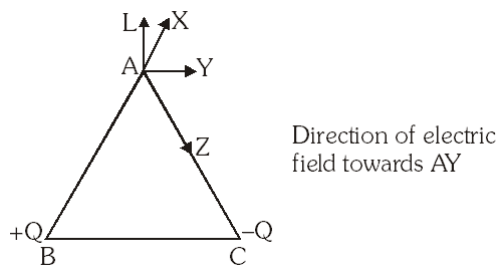
21. **Ans (4)**

$$\begin{aligned} & \text{A} \quad 20\text{cm} \quad \text{B} \quad \leftarrow x \rightarrow \\ & \frac{8 \times 10^{-6}}{K \cdot 8 \times 10^{-6}} = \frac{-2 \times 10^{-6}}{K \cdot 2 \times 10^{-6}} \\ & \frac{(0.20 + x)^2}{2} = \frac{1}{x^2} \\ & \Rightarrow x = 0.2 \text{ m} = 20 \text{ cm} \end{aligned}$$

22. **Ans (1)**

$$\tan \theta = \frac{F_e}{mg} = \frac{kQ^2}{x^2 mg}$$

23. **Ans (2)**



24. **Ans (2)**

By vector addition

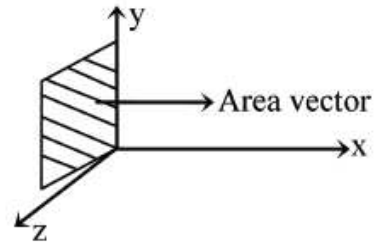
25. **Ans (1)**

The lines proceed from the positive charge and will reach B, if B is negative.

The number of lines/unit area, i.e., the density of lines is greater for the greater charge.

A is +ve, B is -ve and $|A| > |B|$.

26. **Ans (1)**



$$\begin{aligned} \phi &= \mathbf{E} \cdot \mathbf{A} \\ &= (10\hat{i} + 3\hat{j} + 4\hat{k}) \cdot (10\hat{i}) = 100 \text{ units} \end{aligned}$$

27. **Ans (4)**

BY Gauss's law, $\phi_{\text{net}} = 0$

$$\begin{aligned} \text{Thus } \phi_{\text{hemisphere}} &= -\phi_{\text{plane}} = -(-E\pi R^2) \\ &= E(\pi R^2) \end{aligned}$$

28. **Ans (3)**

When charge is placed at corner then flux through cube is

$$\phi = \frac{\Sigma q}{8\epsilon_0} = \frac{q_1 + q_2}{8\epsilon_0}$$

29. **Ans (2)**

$$\phi_{\text{total}} = (\phi_A + \phi_C + \phi_B) \quad (\phi_A = \phi_C)$$

$$\phi_{\text{total}} = (2\phi_A + \phi_B) \quad (\phi_B = \phi)$$

$$\frac{q}{\epsilon_0} = 2\phi_A + \phi \Rightarrow \phi_A = \frac{1}{2} \left(\frac{q}{\epsilon_0} - \phi \right)$$

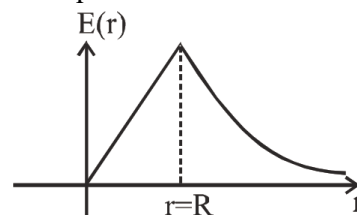
30. **Ans (3)**

For uniformly charged non conducting sphere

$$E = \frac{kQr}{R^2}, \text{ for } r \leq R$$

$$\Rightarrow E \propto r$$

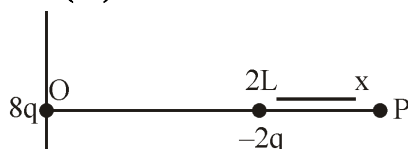
$$E = \frac{kQ}{r^2}, \text{ for } r > R \Rightarrow E \propto \frac{1}{r^2}$$



31. **Ans (1)**

Electric field inside shell is zero.

32. **Ans (2)**



$$X = \frac{2L}{\sqrt{4-1}} = 2L$$

Distance of P for O = 4L

35. **Ans (1)**

$$\phi = \frac{q}{\epsilon_0} = \frac{8.85 \times 10^{-8}}{8.85 \times 10^{-12}} = 10^4 (V - m)$$

SECTION-B

36. **Ans (2)**

$$W_1 = U_f - U_i =$$

$$- \frac{GMm}{R} - \left(- \frac{3}{2} \frac{GMm}{R} \right) = \frac{GMm}{2R}$$

$$W_2 = U_f - U_i = 0 - \left(- \frac{GMm}{R} \right) = \frac{GMm}{R}$$

$$\frac{W_1}{W_2} = \frac{\frac{GMm}{2R}}{\frac{GMm}{R}} = \frac{1}{2}$$

37. **Ans (1)**

$$g_m = \frac{1}{6} g_e$$

$$\frac{4}{3} \pi R_m \rho_m G = \frac{1}{6} \times \frac{4}{3} \pi R_e \rho_e G$$

$$R_m = \frac{\rho_e}{6\rho_m} R_e = \frac{5}{3} \times \frac{1}{6} \times R_e$$

$$R_m = \frac{5}{18} R_e$$

38. **Ans (3)**

$$W_d = \frac{1}{4} \times w$$

$$mg_d = \frac{1}{4} \times mg$$

$$g \left(1 - \frac{d}{R} \right) = \frac{g}{4}$$

$$1 - \frac{d}{R} = \frac{1}{4}$$

$$\frac{d}{R} = \frac{3}{4} \Rightarrow d = \frac{3}{4} R$$

39. **Ans (4)**

$$a = \frac{T_2}{M_1 + M_2}, T_1 = M_1 a$$

$$\therefore T_1 = M_1 \times \frac{T_2}{M_1 + M_2}$$

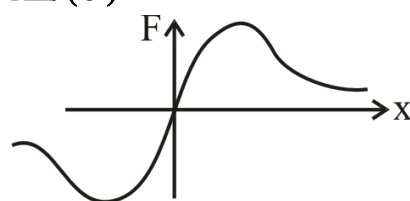
$$\therefore \frac{T_1}{T_2} = \frac{M_1}{M_1 + M_2}$$

40. **Ans (3)**

$$t = \frac{V - U}{a}, \left\{ a = -\frac{5}{5} = -1 \right\}$$

$$t = \frac{0 - 40}{-1} = 40 \text{ sec.}$$

44. **Ans (3)**



45. **Ans (3)**

Electric field will be infinite near both charges.

Since 2Q is the bigger charge hence electric field will be zero near the smaller charge.

46. **Ans (1)**

$$k \frac{\lambda_1}{R} = 2 \frac{k\lambda_2}{R}$$

$$\lambda_1 = 2\lambda_2$$

47. **Ans (1)**

No. of field lines passing through S_1 and S_2 are same

$\therefore$ flux is same

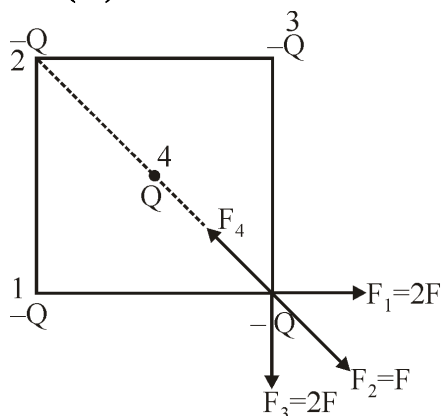
48. **Ans (4)**

$$\phi = \phi_{EFGH} - \phi_{ABCD}$$

$$= b\sqrt{2}a^2 - b\sqrt{a}a^2$$

$$= ba^{5/2} (\sqrt{2} - 1)$$

49. Ans (2)



$$F_1 = F_3 = \frac{kQ^2}{a^2} = 2F$$

$$F_2 = \frac{kQ^2}{2a^2} = F$$

$$F_4 = (2\sqrt{2} + 1)F$$

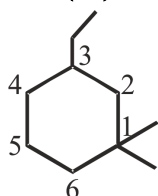
$$\frac{kqQ}{(a/\sqrt{2})^2} = (2\sqrt{2} + 1) \frac{kQ^2}{2a^2}$$

$$a = \frac{(2\sqrt{2} + 1)}{4} Q$$

SUBJECT : CHEMISTRY

SECTION-A

52. Ans (2)



69. Ans (4)

$$r = k[A]^x$$

$$1 \times 10^{-4} = k(0.01)^x \quad \dots\dots(1)$$

$$1.4 \times 10^{-4} = k(0.02)^x \quad \dots\dots(2)$$

$$1 \div 2$$

$$\frac{1}{\sqrt{2}} = \left(\frac{1}{2}\right)^x \Rightarrow \left(\frac{1}{2}\right)^{1/2} = \left(\frac{1}{2}\right)^x$$

$$x = \frac{1}{2}$$

$$r = k[A]^{1/2}$$

71. Ans (4)

Catalyst does not change the value of ΔH of a reaction

→ In absence of catalyst

$$\Delta H = (E_a)_f - (E_a)_i$$

$$\Delta H = 180 - 200$$

$$\Delta H = -20 \text{ KJ mol}^{-1}$$

→ In presence of catalyst

$$\Delta H^1 = (E_a)_f^1 - (E_a)_i^1$$

$$\Delta H^1 = 80 - 100$$

$$\Delta H^1 = -20 \text{ KJ mol}^{-1}$$

$$\Delta H = \Delta H^1$$

72. Ans (3)

Rate constant depends only on nature of reactant, temperature & catalyst.

74. Ans (1)

$$\log \frac{P_2}{P_1} = \frac{\Delta H_{\text{vap}}}{2.303R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$$

$$\log \frac{760}{400} = \frac{\Delta H_{\text{vap}}}{2.303 \times 8.314} \left[\frac{25}{427 \times 452} \right]$$

$$\Delta H_{\text{vap}} \approx 41200 \text{ J mol}^{-1}$$

$$\approx 41.2 \text{ kJ mol}^{-1}$$

75. Ans (2)

A more energetic solvent molecules in solution chamber (high temperature) will resist energetic solvent molecular from entering into solution chamber.

76. Ans (2)

$$N = \frac{\text{gram equivalents}}{\text{volume}}$$

$$= \frac{2}{40} \times \frac{1}{0.1} = \frac{1}{2} = 0.5 \text{ N}$$

77. Ans (3)

$$\frac{-\Delta[A]}{\Delta t} = \frac{1}{2} \left(-\frac{\Delta[B]}{\Delta t} \right)$$

$$\therefore -\frac{\Delta[B]}{\Delta t} = 2 \times 2.6 \times 10^{-2} = 5.2 \times 10^{-2} \text{ Ms}^{-1}$$

80. **Ans (2)**

$$x = kt = 0.025 \times 15 = 0.375 \text{ M}$$

$$[A]_t = 0.6 - 0.375$$

$$= 0.225 \text{ M}$$

81. **Ans (3)**

$$r = k[A]^{1/2} [B]^{3/2} [C]^{-1}$$

$$\text{order} = \frac{1}{2} + \frac{3}{2} - 1 = 1$$

$$\text{unit of } K = \text{time}^{-1}$$

82. **Ans (4)**

$$t_{93.75\%} = 4t_{50\%}$$

$$= 4 \times 120 = 480 \text{ hr.}$$

$$= 480 \times 60 = 28,800 \text{ min.}$$

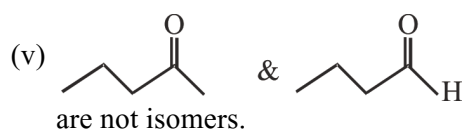
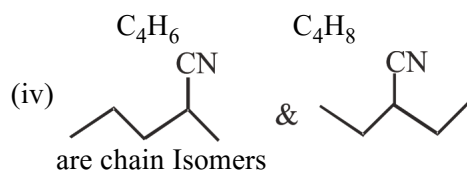
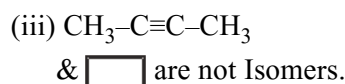
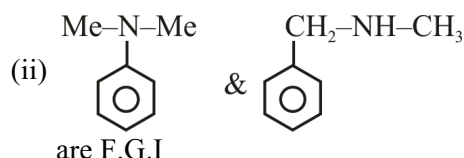
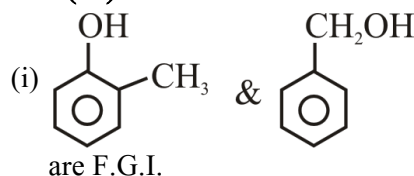
83. **Ans (3)**

$$\%S = \frac{W_B}{V \text{ (mL)}} \times 100$$

$$= \frac{2}{300} \times 100 = 0.66\%$$

SECTION-B

89. **Ans (1)**



95. **Ans (2)**

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$$

$$\log \frac{k_2}{k_1} = \frac{180 \times 10^3}{2.303 \times 8.314} \left[\frac{1}{300} - \frac{1}{310} \right]$$

$$\log \frac{k_2}{k_1} = 1$$

$$\frac{k_2}{k_1} = 10 \Rightarrow k_2 = 10 k_1$$

Rate of proportional to rate constant, rate will increase by a factor of 10.

98. **Ans (3)**

$$\log \frac{K_2}{K_1} = \frac{E_a}{2.303R} \left(\frac{T_2 - T_1}{T_1 T_2} \right)$$

$$\therefore E_a = 0 \therefore \log \frac{K_2}{K_1} = 0$$

$$\text{and } \frac{K_2}{K_1} = 1, K_2 = K_1 = 1 \times 10^{-6} \text{ s}^{-1}$$

99. **Ans (3)**

$$t \propto a^{1-n}$$

$$\left(\frac{t_1}{t_2} \right) = \left(\frac{a_1}{a_2} \right)^{1-n}$$

$$\frac{200}{40} = \left(\frac{0.1}{0.5} \right)^{1-n}$$

$$(5)^1 = \left(\frac{1}{5} \right)^{1-n}$$

$$\left(\frac{1}{5} \right)^{-1} = \left(\frac{1}{5} \right)^{1-n}$$

$$1 - n = -1$$

$$n = 2$$

100. **Ans (4)**

$$\text{Molarity (M)} = \frac{6.8/34}{224/1000} = \frac{200}{224} \text{ M}$$

$$\text{Volume strength of } \text{H}_2\text{O}_2 = M \times 11.2$$

$$= \frac{200}{224} \times 11.2 = 10 \text{ V}$$

SUBJECT : BIOLOGY-I

SECTION-A

112. **Ans (2)**

NCERT-XII, Pg. # 50, 51

113. **Ans (2)**

NCERT (XIIth) Pg. # 47

114. **Ans (1)**

Module-1, Pg. # 53

115. **Ans (3)**

NCERT-XII, Pg. # 46, para-3(E),
Pg. # 49, para-5(H)

116. **Ans (3)**

NCERT Pg. # 54, Para-3.6

117. **Ans (2)**

NCERT XII, Pg # 48, fig. 3.6

118. **Ans (2)**

NCERT XII Pg. # 46, Fig.no. 3.1(अ)

120. **Ans (3)**

NCERT Pg.# 43

121. **Ans (4)**

NCERT Pg. # 53, Para- 3.5

122. **Ans (2)**

NCERT-XII, Pg. # 49(E), 53(H), Fig. 3.8(b)

123. **Ans (3)**

NCERT Pg.# 43

124. **Ans (3)**

Module-1, Pg # 49

125. **Ans (3)**

1 primary spermatocyte produces 4 sperms
while 1 primary oocyte produces 1 ovum.

126. **Ans (3)**

Progesterone is responsible for maintenance of
endometrium.

127. **Ans (4)**

NCERT XII Pg. N. 46

128. **Ans (4)**

Module-1, Pg. # 21

129. **Ans (2)**

NCERT-XII, Pg. # 236

130. **Ans (4)**

NCERT-XII, Pg. # 236

131. **Ans (3)**

NCERT-XII, Pg. # 235

132. **Ans (3)**

NCERT-XII, Pg. # 233

135. **Ans (4)**

NCERT-XII, Pg. # 251

SECTION-B

142. **Ans (1)**

NCERT Pg. # 51

143. **Ans (2)**

NCERT(XII) Page#50 Fig:3.9

144. **Ans (2)**

NCERT-XII, Page No- 48

145. **Ans (3)**

NCERT XII Pg # 53

146. **Ans (4)**

NCERT XII, Pg # 47, 2nd para

147. **Ans (4)**

NCERT XII Pg. # 57/64(H)Para:4.1

148. **Ans (2)**

NCERT-XII, Pg. # 235

149. **Ans (3)**

NCERT-XII, Pg. # 233

150. **Ans (4)**

NCERT-XII Pg. # 232 table 13.1

SUBJECT : BIOLOGY-II

SECTION-A

151. **Ans (3)**

NCERT-XII Pg. # 236

152. **Ans (4)**

NCERT-XII Pg. # 237

153. **Ans (2)**

NCERT-XII Pg. # 237

155. **Ans (3)**

NCERT-XII Pg. # 234

157. **Ans (2)**

NCERT-XII, Pg. # 232

158. **Ans (2)**

NCERT-XII, Pg. # 250

160. **Ans (2)**

NCERT-XII Pg. # 233

175. **Ans (1)**

NCERT XII Pg # 220, 235, 237

176. **Ans (4)**

NCERT Pg. # 248

178. **Ans (3)**

Allen Module - 21-23

179. **Ans (2)**

NCERT -XII Pg. No. 168 – 169

180. **Ans (4)**

NCERT XII Bio-167

181. **Ans (1)**

NCERT (XII) Pg# 167/182(H) para: 9.1.2

182. **Ans (1)**

NCERT (XII) Pg # (E) 168 (H) 182 para 9.1.2

183. **Ans (2)**

NCERT (XII) Pg. # 166, para 9.1

184. **Ans (3)**

NCERT(XII) Pg#168 Para:9.1.2

185. **Ans (2)**

NCERT (XII) Pg. # 168,169, Para-9.1.2

SECTION-B

186. **Ans (4)**

NCERT-XII, Pg. # 233

187. **Ans (1)**

NCERT-XII Pg. # 235-236

188. **Ans (2)**

NCERT-XII Pg. # 250

195. **Ans (3)**

NCERT-XII ; Page # 243

198. **Ans (4)**

NCERT XII Page # 168 ; Para # 9.1.2

199. **Ans (3)**

NCERT (XIIth) Pg. # 167 (9.1.2)

200. **Ans (1)**

NCERT(XII) Page# 170/184(H) para 9.1.4